

Sending signals

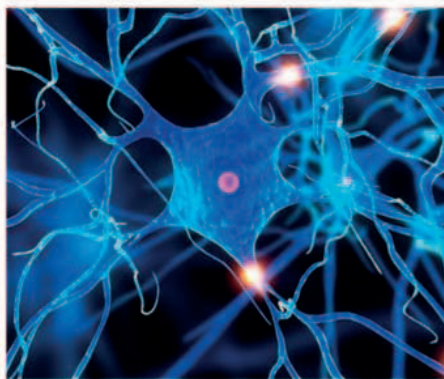
The nervous system is made up of billions of cells called neurons (nerve cells). These are connected to each other by thin, wirelike threads that carry electrical signals and can extend for more than a meter. Neurons form the body's living wiring system and make up most of the cells in your brain.

MAKING ELECTRICITY

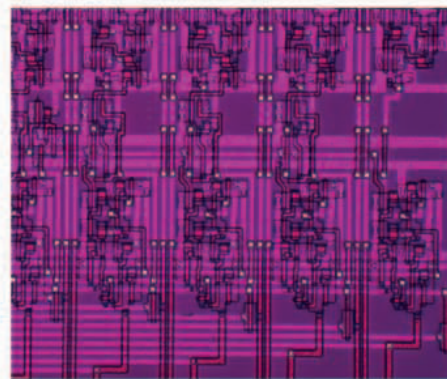
Neurons generate the electrical signals that dart around your nervous system. When a cell is resting (switched off), it builds up a charge. When a signal arrives and switches it on, this charge rushes through the cell to the next neuron.

A LIVING COMPUTER

You may not realize it, but your brain has more processing power than the most advanced computer on Earth. Brain cells work like the tiny transistors that make up a computer chip. But while a computer chip has around a billion transistors with three to four connections each, your brain contains 100 billion neurons, each with up to 10,000 connections to other neurons. The vast number of connections and circuits allows the brain to process millions of pieces of information at the same time.



▲ **BRAIN CIRCUIT** *Unlike a computer, a living brain can grow new connections and continually rewires itself throughout life.*



▲ **COMPUTER CHIP** *The circuits on a computer chip are printed in a factory and cannot change afterward.*